

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-12 10:00 UTC

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Generated 2026-05-12 10:00 UTC

- Cycles recorded: **653**
- Time span: 444.3h (~1.47 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	616
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	35
Schur-Weyl Duality	22
Representation Theory of Symmetric Groups	17
Matroid Theory	16
Free Probability	14
Additive Combinatorics	10
Free Probability Theory	10
Noncommutative L^p Geometry	9
Polymatroid Theory	8
Algebraic Geometry	7
Random Matrix Theory	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4

Field	Cycles
Persistent Homology	4
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Diophantine Approximation	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-12 03:25 UTC	INCONCLUSIVE	Free Cumulant Sum Bounded by Log-Size for Read-Twice BPs
2026-05-12 04:34 UTC	INCONCLUSIVE	Condition Number of Moment Matrix Inversely Proportional to SOS D
2026-05-12 04:40 UTC	INCONCLUSIVE	Projective Plane Incidence Bounds Nisan-Wigderson Seed Length
2026-05-12 05:17 UTC	INCONCLUSIVE	Fourier Coefficient Spread and ACC ⁰ Circuit Lower Bounds
2026-05-12 05:43 UTC	INCONCLUSIVE	Fourier Coefficient Decay Bounds AC ⁰ Circuit Size for Parity-Inse
2026-05-12 06:36 UTC	INCONCLUSIVE	Invariant Ring Degree Bounds SOS Refutation Degree for Symmetric
2026-05-12 06:43 UTC	INCONCLUSIVE	Noncommutative L ^p Norm of Communication Matrix Bounds Randomized
2026-05-12 07:48 UTC	INCONCLUSIVE	Matroid Rank Inverse Proportional to Monotone DNF Size for k-CLIQ
2026-05-12 07:54 UTC	INCONCLUSIVE	Matroid Rank Submodularity in Monotone DNF for k-CLIQUE
2026-05-12 08:01 UTC	INCONCLUSIVE	Tensor Product Multiplicity Inverse Proportional to Monotone Circ
2026-05-12 08:14 UTC	INCONCLUSIVE	Persistent Homology of Read-Twice BPs Bounded by Log-Size

Time	Verdict	Title
2026-05-12 08:22 UTC	INCONCLUSIVE	Hilbert Function Growth Inversely Proportional to Extended Frege
2026-05-12 08:32 UTC	INCONCLUSIVE	Block Design Discrepancy and Communication Complexity Lower Bound
2026-05-12 08:44 UTC	INCONCLUSIVE	Second Eigenvalue Inverse Proportional to Resolution Proof Length
2026-05-12 09:26 UTC	INCONCLUSIVE	Real Radical Dimension Bounds SOS Degree for Max-CUT

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```